

UNIVERSITY OF SINDH, JAMSHORO
Institute of Mathematics & Computer Science (IMCS)
Department of Computer Science

Assignment No. 1

Artificial Intelligence
PEAS Framework & Task Environment Analysis

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Subject	Artificial Intelligence (AI)
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System Selected: **A — Automated Warehouse Sorting Robot** (e.g., Amazon Robotics)

Task 1: PEAS Specification

Performance Measures

1. **Sorting Accuracy Rate:** Percentage of packages correctly identified, sorted, and routed to the right bin/conveyor (target > 99.5%).
2. **Throughput (Packages per Hour):** Number of packages successfully processed per hour, measured against the facility's target throughput.
3. **Collision and Damage Rate:** Number of package drops, robot collisions, or product damage incidents per 1,000 packages handled.
4. **Idle Time Percentage:** Fraction of operational time where the robot is not actively sorting — minimized to maximize efficiency.
5. **Battery/Energy Efficiency:** Packages sorted per kWh consumed, balancing performance against operating cost.

Environment

The robot operates inside a large enclosed warehouse facility. The environment includes moving conveyor belts, static shelving units, and dynamic obstacles such as human workers, forklifts, and other robots. Packages vary in size, weight, shape, and label orientation. Lighting is controlled but can vary near loading docks.

Actuators

1. **Robotic Arm with Multi-DOF Gripper:** Picks up packages of varying sizes and orientations.
2. **Omnidirectional Drive Wheels:** Allows movement in any direction across the warehouse floor.
3. **Conveyor Belt Interface Controller:** Routes packages onto the correct outbound conveyor lane.
4. **Digital Label Printer/Applicator:** Re-labels packages when barcodes are damaged or missing.
5. **Alert Speaker and Warning Light:** Signals nearby humans of robot movement for safety.

Sensors

1. **High-Resolution RGB Camera (4K):** Reads barcodes, QR codes, and package labels.
2. **3D LiDAR Scanner:** Maps the warehouse environment and detects obstacles in real time.
3. **Infrared Depth Sensor (e.g., Intel RealSense):** Measures package dimensions and detects irregular shapes.
4. **Force/Torque Sensor on Gripper:** Detects grip strength and prevents package damage.

5. **IMU (Inertial Measurement Unit):** Tracks robot orientation and detects tipping.
6. **RFID Reader:** Reads embedded RFID tags on pallets and high-value packages.

Task 2: Environment Classification

Fully Observable vs. Partially Observable

Choice: Partially Observable

Justification: The robot cannot see all areas of the warehouse simultaneously — packages behind shelves, items on far conveyors, and occluded objects are invisible at any given moment.

Single-agent vs. Multi-agent

Choice: Multi-agent

Justification: Multiple sorting robots, forklifts, and human workers operate concurrently in the same space, requiring coordination and collision avoidance.

Deterministic vs. Stochastic

Choice: Stochastic

Justification: Package arrival times, weights, and orientations are unpredictable; human workers introduce random movement — outcomes cannot be perfectly predicted from actions alone.

Episodic vs. Sequential

Choice: Sequential

Justification: Decisions are interdependent — choosing a path to avoid one robot affects future positioning, and misrouting one package causes downstream congestion.

Static vs. Dynamic

Choice: Dynamic

Justification: The environment changes continuously while the robot deliberates: new packages arrive, other robots move, and workers cross paths without waiting for the robot's action.

Discrete vs. Continuous

Choice: Continuous

Justification: The robot moves through continuous physical space, gripper forces are real-valued, and package positions are not confined to a fixed grid.

Task 3: Critical Analysis

1. Greatest Technical Challenge

The most technically challenging dimension is the **Dynamic** nature of the environment. A live warehouse changes every millisecond — human workers move unpredictably, robots re-route due to obstacles, and new packages are continuously added to conveyors. This demands real-time perception, planning, and replanning at very high frequency. The AI engineer must implement low-latency sensor fusion, predictive path planning (e.g., RRT* or model-predictive control), and multi-agent communication protocols — all while ensuring safety guarantees. No pre-computed plan remains valid for long, making dynamic re-planning the core engineering bottleneck.

2. Utility Function: Speed vs. Safety Trade-off

A meaningful trade-off is between **operational speed** (throughput) and **safety** (avoiding collisions and damage). The proposed utility function is:

$$U = w_1 \times (\text{packages_per_hour} / \text{max_throughput}) - w_2 \times (\text{collision_risk_score}) - w_3 \times (\text{damage_rate})$$

With default weights $w_1 = 0.5$, $w_2 = 0.3$, $w_3 = 0.2$, the robot balances speed and safety moderately. If weight w_2 (collision risk) were **doubled to 0.6**, the robot would adopt a significantly more conservative movement strategy — slowing near humans, maintaining larger safety margins, and yielding more frequently to other agents. Throughput drops noticeably, but incidents and liability are minimized — a desirable trade-off in high-density facilities with frequent human presence.

Task 4: Structured Data Design

Structured Representation (JSON)

```
{
  "system_name": "Automated Warehouse Sorting Robot",
  "peas": {
    "performance_measures": [
      "sorting_accuracy_rate (target > 99.5%)",
      "throughput_packages_per_hour",
      "collision_and_damage_rate_per_1000_packages",
      "idle_time_percentage",
      "battery_energy_efficiency_packages_per_kWh"
    ],
    "environment": "Enclosed warehouse with conveyors, workers, forklifts, robots",
    "actuators": [
      "robotic_arm_with_multi_DOF_gripper",
      "omnidirectional_drive_wheels",
      "conveyor_belt_interface_controller",
      "digital_label_printer_applicator",
      "alert_speaker_and_warning_light"
    ]
  }
}
```

```

],
"sensors": [
  "high_resolution_RGB_camera_4K",
  "3D_LiDAR_scanner",
  "infrared_depth_sensor_RealSense",
  "force_torque_sensor_on_gripper",
  "IMU_inertial_measurement_unit",
  "RFID_reader"
]
},
"environment_classification": {
  "observability": {
    "choice": "Partially Observable",
    "justification": "Cannot see occluded packages or far conveyor areas at once"
  },
  "agents": {
    "choice": "Multi-agent",
    "justification": "Multiple robots, forklifts, and humans operate concurrently"
  },
  "determinism": {
    "choice": "Stochastic",
    "justification": "Package arrivals and worker movements are unpredictable"
  },
  "episodicity": {
    "choice": "Sequential",
    "justification": "Routing decisions affect future states and conveyor flow"
  },
  "dynamism": {
    "choice": "Dynamic",
    "justification": "Environment changes continuously without waiting for robot"
  },
  "continuity": {
    "choice": "Continuous",
    "justification": "Physical movement and gripper forces are real-valued"
  }
},
"utility_function":
  "U = 0.5*(packages_per_hour/max_throughput)
    - 0.3*(collision_risk_score) - 0.2*(damage_rate)"
}

```